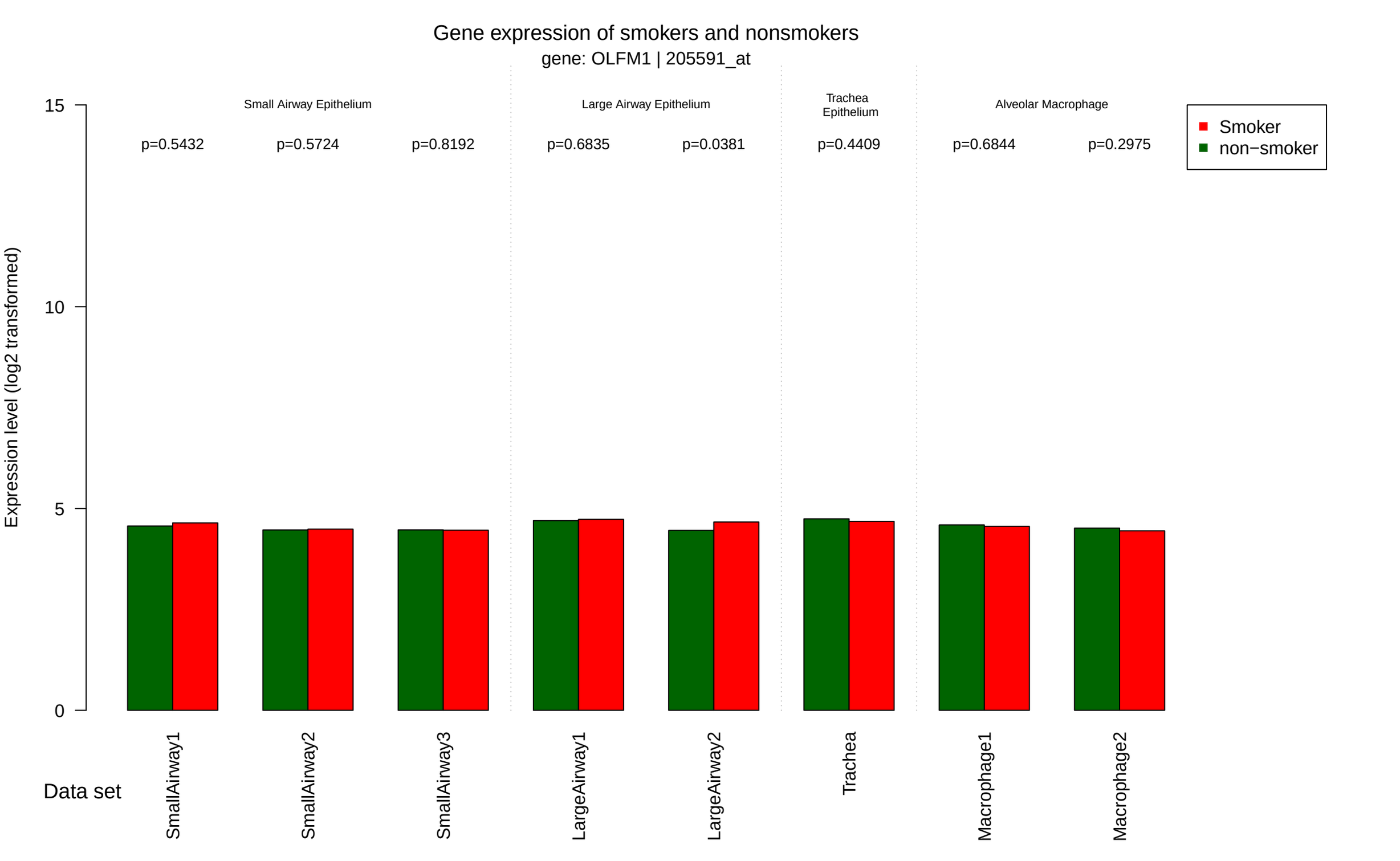
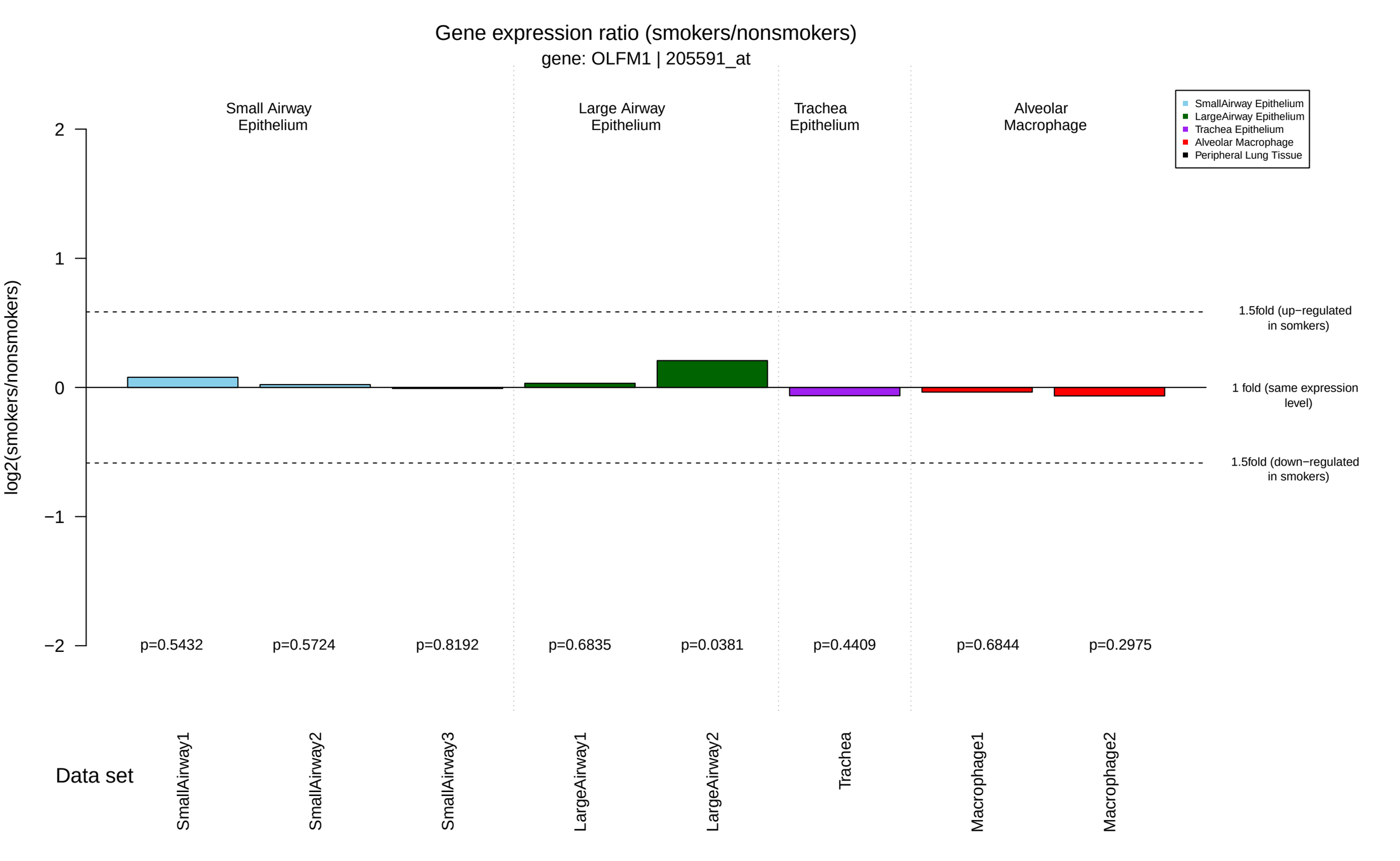
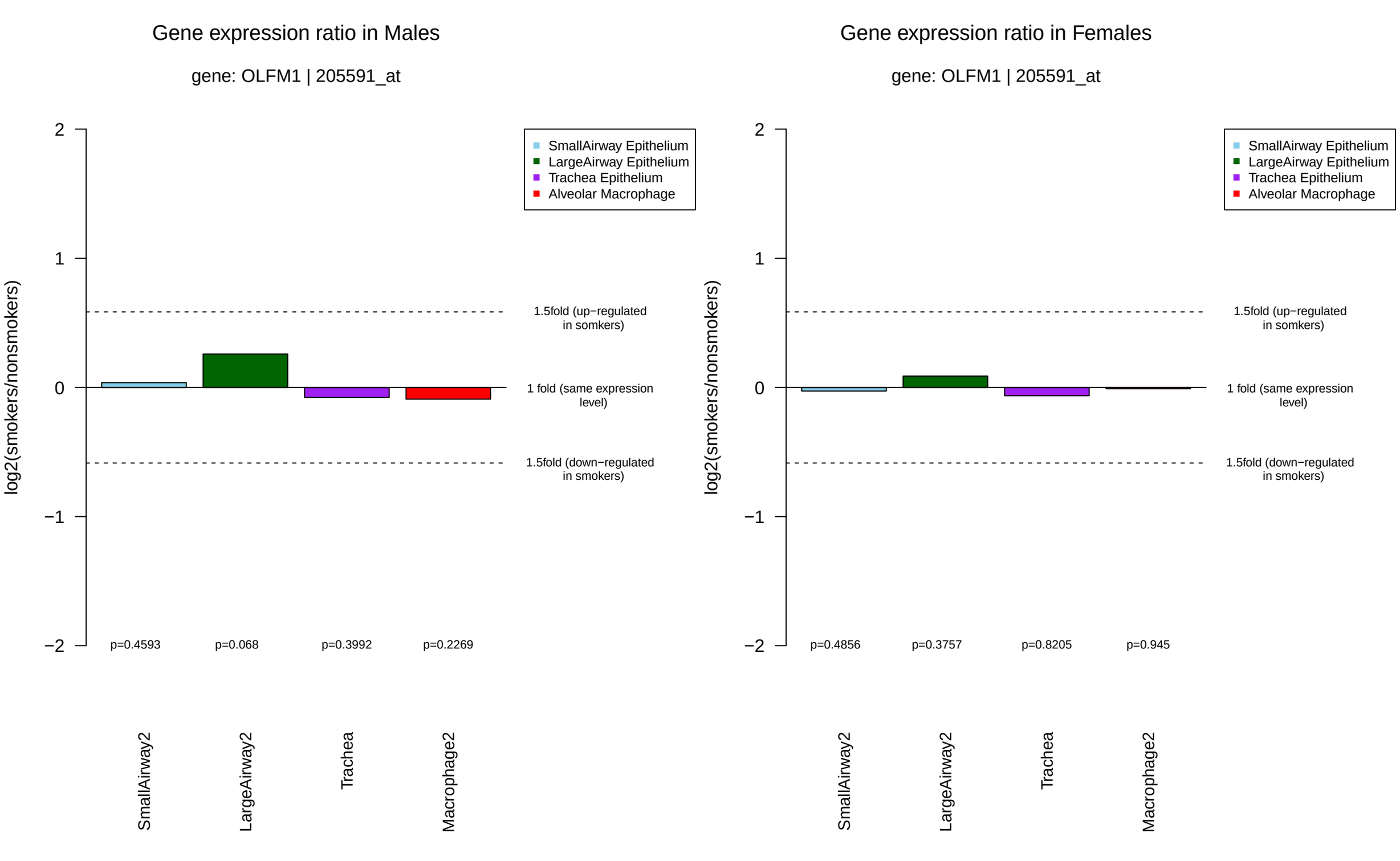
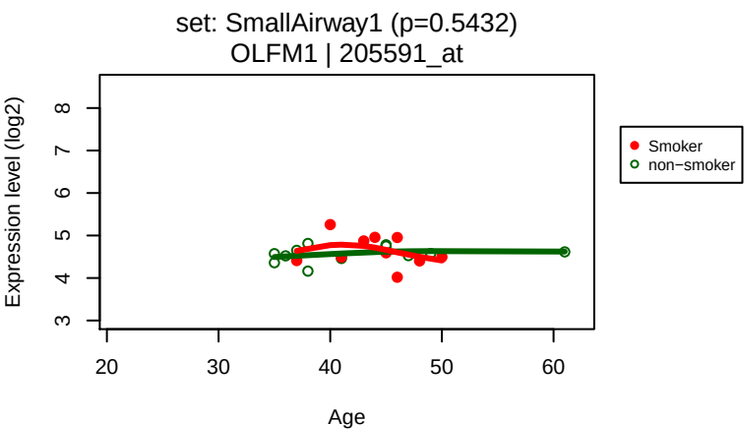
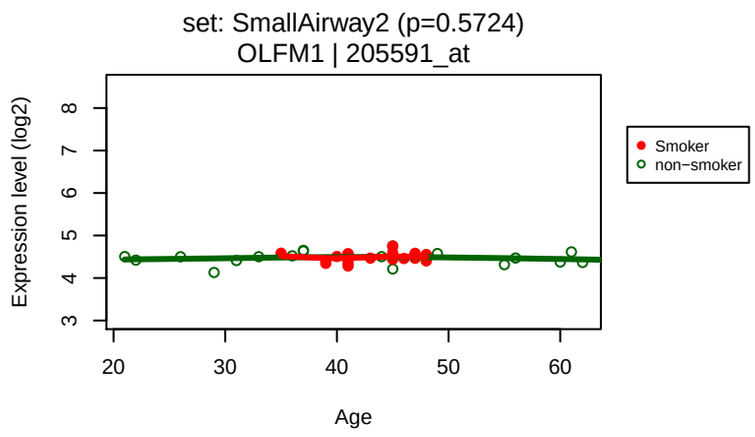
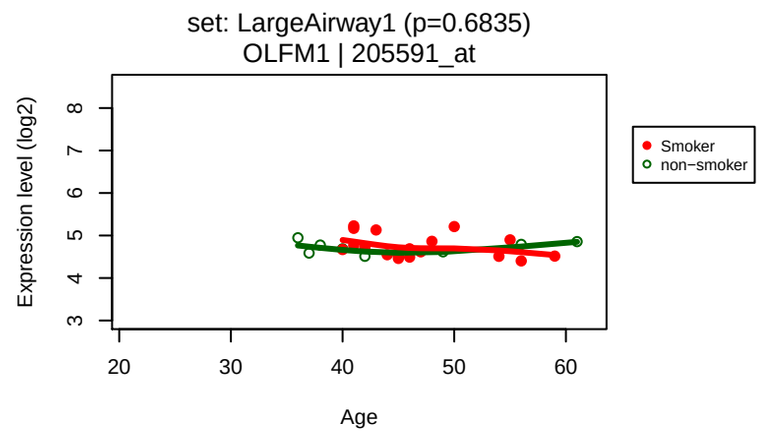

Gene expression of smokers and nonsmokers
gene: OLFM1 | 205591_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.5432
p=0.5724
p=0.8192
p=0.6835
p=0.0381
p=0.4409
p=0.6844
p=0.2975
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

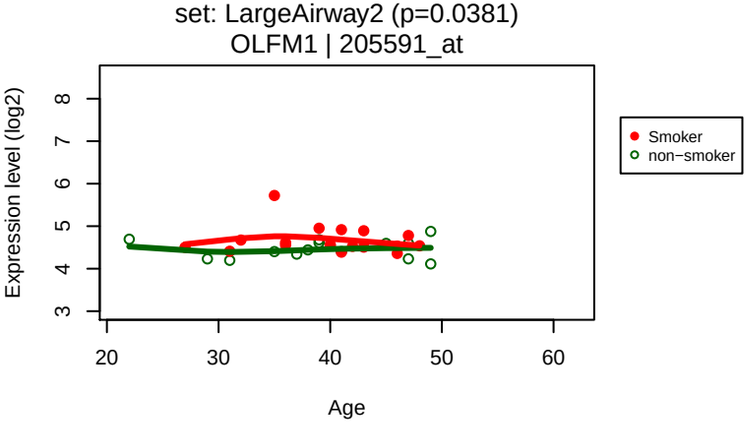

Gene expression ratio (smokers/nonsmokers)
gene: OLFM1 | 205591_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5432
p=0.5724
p=0.8192
p=0.6835
p=0.0381
p=0.4409
p=0.6844
p=0.2975
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

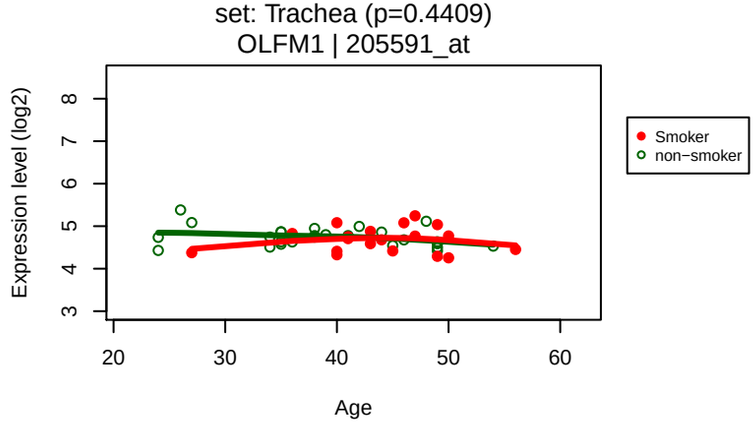

Gene expression ratio in Males
gene: OLFM1 | 205591_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4593
p=0.068
p=0.3992
p=0.2269
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: OLFM1 | 205591_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4856
p=0.3757
p=0.8205
p=0.945
SmallAirway2
LargeAirway2
Trachea
Macrophage2

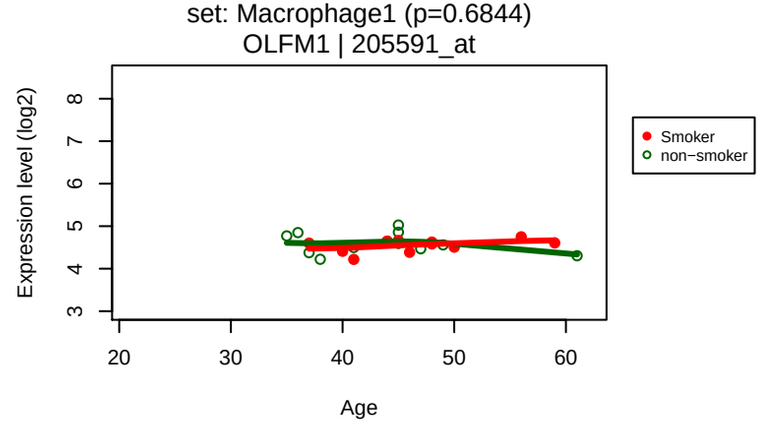

set: SmallAirway1 (p=0.5432)
OLFM1 | 205591_at
Expression level (log2)
Age
Smoker
non−smoker

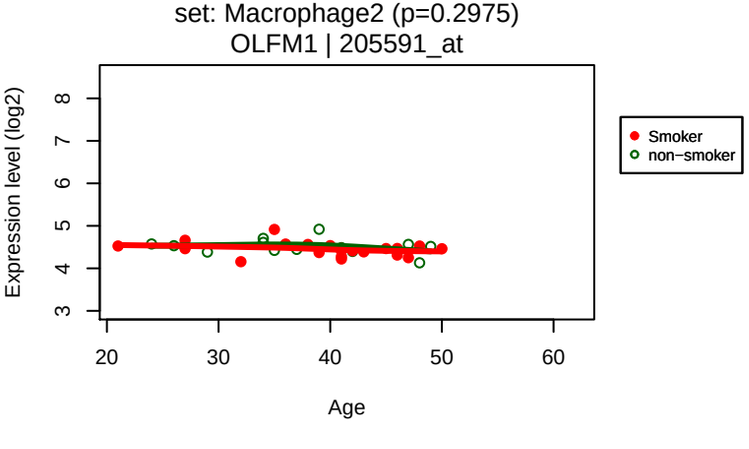

set: SmallAirway2 (p=0.5724)
OLFM1 | 205591_at
Expression level (log2)
Age
Smoker
non−smoker

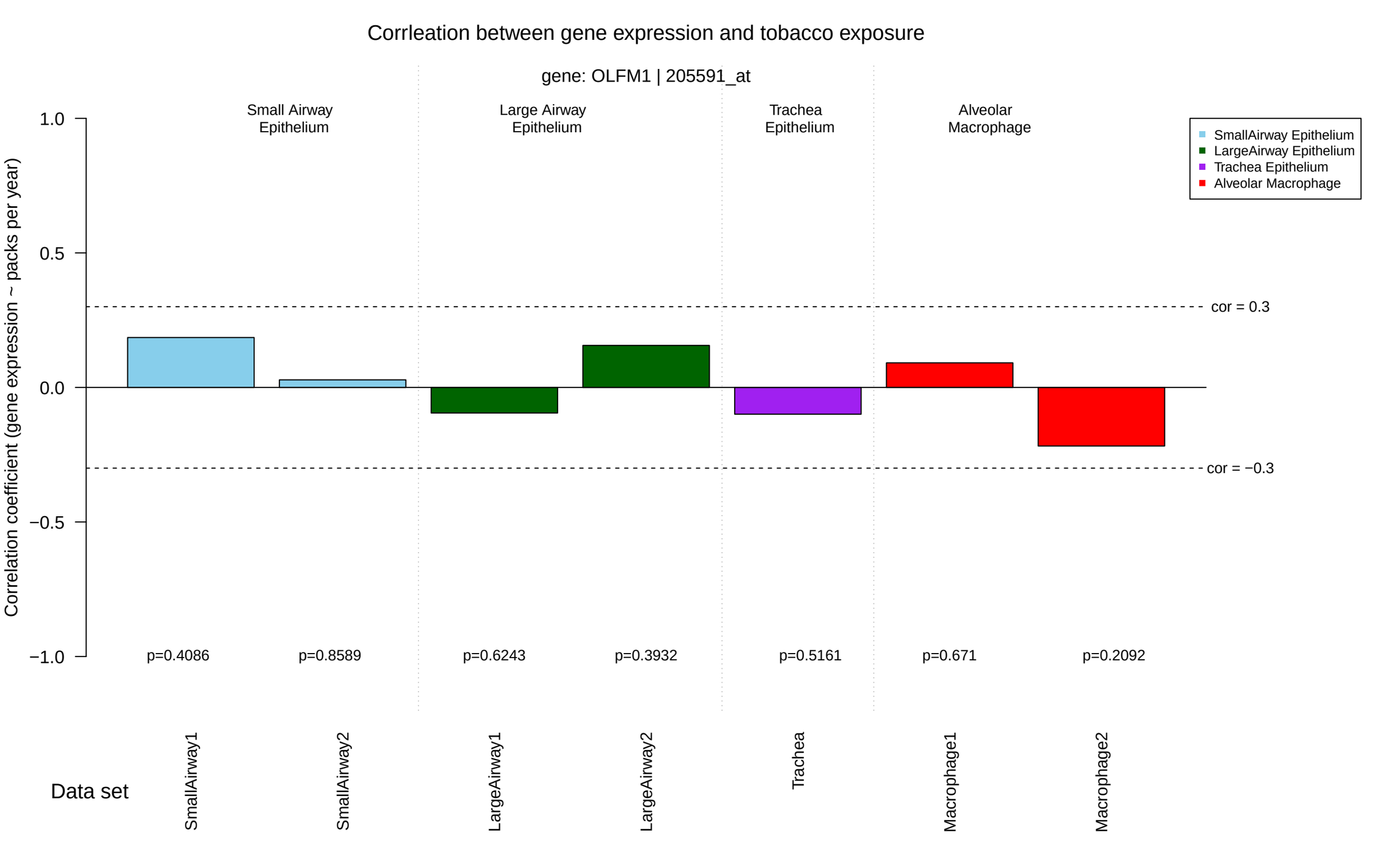

set: LargeAirway1 (p=0.6835)
OLFM1 | 205591_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0381)
OLFM1 | 205591_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.4409)
OLFM1 | 205591_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.6844)
OLFM1 | 205591_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.2975)
OLFM1 | 205591_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: OLFM1 | 205591_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.4086
p=0.8589
p=0.6243
p=0.3932
p=0.5161
p=0.671
p=0.2092
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2